

Nios[®] V/m Processor with DMA and OCM

Agilex[™] 5 FPGA

Last updated: **2024-09-30**

Contents

1.0	Theory of Operation	3
1.1	IP Cores	3
2.0	Simulating the design	4
3.0	Executing the design on Development Kit.....	5
3.1	Creating and validating the design	5
4.0	Expected Results	6
5.0	Document Revision History	7

1.0 Theory of Operation

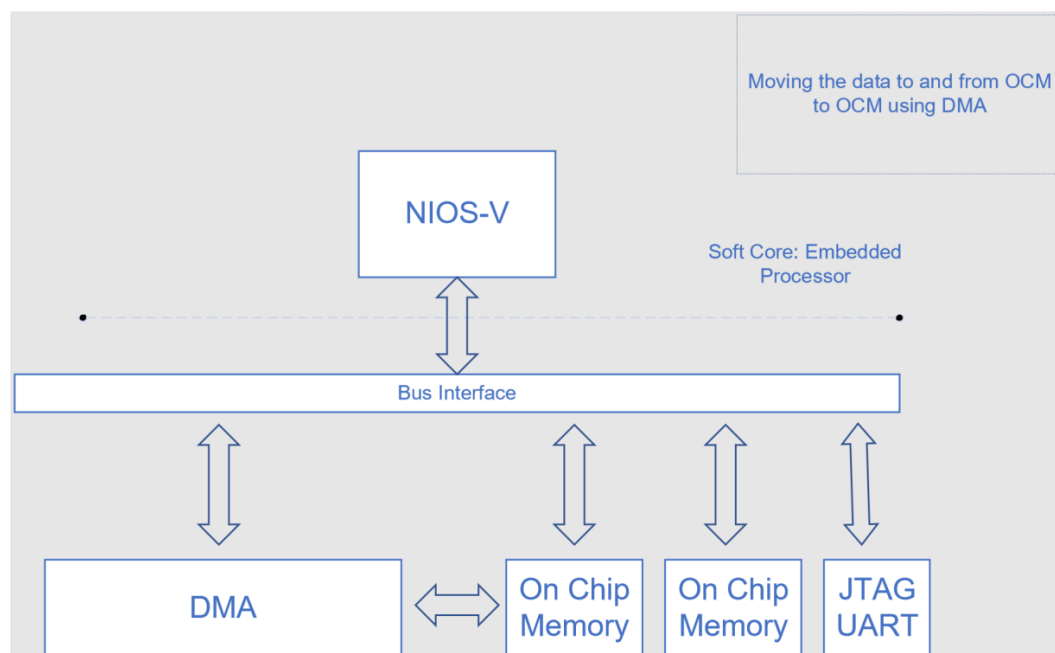
This design demonstrates the transaction between the Nios® V processor with DMA and OCM core for the Agilex™ 5 FPGA E-Series 065B Premium Development Kit (ES1) DK-A5E065BB32AES1.

This example design includes a Nios® V processor connected to the DMA, On Chip RAM-II and JTAG UART IP.

The objective of the design is to accomplish a data transfer between the 2 On Chip RAM-II using a DMA (MSGDMA) IP.

DMA facilitates the data transfer which is then read back by the processor.

Figure 1-1. Block Diagram



1.1 IP Cores

The following IPs are used in this design.

- Nios V/m soft processor core
- On Chip RAM-II
- JTAG UART
- MSGDMA
- System ID

2.0 Simulating the design

Note: Please refer to the readme.txt file in the package for the steps to create the design, application and generate the programming files.

- Unpackage/extract the design in your working directory.
- Simulate the design using Quartus Prime Pro tool.

3.0 Executing the design on Development Kit

3.1 Creating and validating the design

Note: Please refer to the readme.txt file in the package for the steps to create the design, application and generate the programming files.

- Unpackage/extract the design in your working directory.
Locate the "ready_to_test" folder within the package.
- The folder contains the necessary files for executing the application on the board.
Refer to the readme file for the steps to program the application files on the board.
- Validate the design by observing the prints on the terminal.

4.0 Expected Results

The following is the output as observed on the JTAG UART terminal. The output is analogous to the logic from the application code. Users should be able to observe same output on their terminal/setup.

```
# Hello World from NIOSV/m core on SM !
# *****Run DMA Test - OCM to OCM *****
# writing into onchip memory from NIOS V
# Addr 0x200000 Data written 0x0
# Addr 0x200004 Data written 0x4
# Addr 0x200008 Data written 0x8
# Addr 0x20000c Data written 0xc
# Addr 0x200010 Data written 0x10
# Addr 0x200014 Data written 0x14
# Addr 0x200018 Data written 0x18
# Addr 0x20001c Data written 0x1c
# Data write to onchip memory completed
# DMA is open mSG-DMA
# Created descriptor definition for mSG-DMA
# status is 0
# Addr 0x120000 Data read 0x0
# base 0x120000 offset_add 0x0 ,expected 0x0 read 0x0
# Data matched !
# Addr 0x120004 Data read 0x4
# base 0x120000 offset_add 0x4 ,expected 0x4 read 0x4
# Data matched !
# Addr 0x120008 Data read 0x8
# base 0x120000 offset_add 0x8 ,expected 0x8 read 0x8
# Data matched !
# Addr 0x12000c Data read 0xc
# base 0x120000 offset_add 0xc ,expected 0xc read 0xc
# Data matched !
# Addr 0x120010 Data read 0x10
# base 0x120000 offset_add 0x10 ,expected 0x10 read 0x10
# Data matched !
# Addr 0x120014 Data read 0x14
# base 0x120000 offset_add 0x14 ,expected 0x14 read 0x14
# Data matched !
# Addr 0x120018 Data read 0x18
# base 0x120000 offset_add 0x18 ,expected 0x18 read 0x18
# Data matched !
# Addr 0x12001c Data read 0x1c
# base 0x120000 offset_add 0x1c ,expected 0x1c read 0x1c
# Data matched !
# Memory copy over DMA was a SUCCESS. Session will now close.
# Print the value of System ID
# System ID from Peripheral core is 0xA5
```

5.0 Document Revision History

Date	Version	Changes
2024-09-30	24.3.0	Initial release on github

© Altera Corporation. Altera, the Altera logo, the 'a' logo, and other Altera marks are trademarks of Altera Corporation. Altera and Intel warrant performance of its FPGA and semiconductor products to current specifications in accordance with Altera's or Intel's standard warranty as applicable but reserves the right to make changes to any products and services at any time without notice. Altera and Intel assume no responsibility or liability arising out of the application or use of any information, product, or service described herein except as expressly agreed to in writing by Altera or Intel. Altera and Intel customers are advised to obtain the latest version of device specifications before relying on any published information and before placing orders for products or services.

*Other names and brands may be claimed as the property of others.